

EVA: A 16mm² 1.54TFLOPS Tiled-Based Accelerator for Evolvable Edge Computing

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Abstract

EVA is an accelerator designed for evolvable edge computing applications. It integrates programmable processing array tiles (A-Tiles) for performance and utilization, and RISC-V CPU tiles (C-Tiles) for control and quick reconfiguration at a 100ns timescale. EVA features a versatile interconnect fabric, a distributed memory system, and flexible mapping. A 16mm² EVA test chip achieves a throughput of 1.54TFLOPS (FP16) at 0.9V and a peak energy efficiency of 708GFLOPS/W at 0.49V, demonstrating high compute density and efficiency for diverse workloads.

Introduction

Many edge computing tasks, including machine learning, control, and communication DSP, often rely on a set of heterogeneous compute kernels. These tasks can evolve rapidly, altering the use of the kernels at runtime (Fig. 1(a)). Programmable and reconfigurable hardware addresses this need. Amber [1], a coarse-grained reconfigurable architecture (CGRA) with an array of processing elements (PEs), supports various kernels with flexible reconfiguration but relies on less agile bitstream reconfiguration. DAP [2], a domain-specific accelerator for DSP, features heterogeneous compute engines but may face under-utilization and inefficiency as tasks evolve and kernels change.

We present EVA, an accelerator for evolvable workloads. EVA (Fig. 1(b)) combines PE array tiles (A-Tiles) with RISC-V CPU tiles (C-Tiles) (Fig. 2). The programmable PEs in A-Tiles support diverse compute kernels with high utilization, while the embedded C-Tiles provide extended processing and facilitate rapid runtime adaptation. EVA's versatile interconnect fabric, including local systolic interconnects, tile-level router networks, and inter-tile AXI buses, enables efficient data movement. Aided by a compiler, EVA supports flexible spatial and temporal workload mapping options to optimize performance and energy.

Tile-Based Architecture and Runtime Adaptation

EVA's architecture consists of 4 quadrants connected to the center C-Tile acting as the global controller (Fig. 2(a)). Each quadrant contains a quadrant C-Tile and 4 A-Tiles (Fig. 2(b)). Each A-Tile includes a 64-PE array and 16-KB of tile memory (Fig. 3(a)). The programmable PEs support a set of operations (Fig. 3(c)), allowing the A-Tile to handle various kernels with high utilization. This design, with programmable PEs and quadrant C-Tile scheduling of A-Tile programs, enables efficient reuse of kernels on A-Tiles.

The PE execution pipeline (Fig. 3(b)) begins by fetching an instruction from an 8x16-bit instruction register file (IRF), decoding it, and sending it to an FP16 floating-point unit (FPU). Data hazards are detected and prevented by data forwarding. A write arbiter decides whether results are written to the 8x16-bit data register file (DRF) or the tile memory. The PE instruction set architecture (ISA) (Fig. 3(c)) supports scalar computation, data movement, and conditional decisions, all encoded in 16-bit instructions. The processing array is homogeneous, except for the 8 left-most PEs in the array, which include division and square-root units. Complex-number operations can be decomposed into scalar operations and programmed on a single PE or distributed to multiple PEs (Fig. 4).

EVA's runtime adaptation is enabled through instructions in the A-Tile PEs, and via scheduling and reconfiguration by the quadrant C-Tiles (Fig. 5). Programs loaded in the PE IRF allow limited adaptation at the nanosecond timescale. The quadrant C-Tiles can schedule these PE programs at a 10ns timescale and reconfigure the PEs by updating their instructions at a 100ns timescale (75ns at 1.5GHz). Each quadrant can be independently controlled by its quadrant C-Tile, coordinated by the center C-Tile.

Distributed Memory and Versatile Interconnect Fabric

EVA features distributed local memories and a versatile interconnect fabric. Within an A-Tile, PEs use systolic interconnects for local data movement and a router network for distant communication (Fig. 6(a)). Besides data movement, the router network also delivers new

instructions from C-Tiles to reconfigure the PEs in a parallel and distributed manner, saving 10.5% in area for a dedicated distribution network. Controlled by PE instructions, these interconnects provide up to 36GB/s bandwidth per PE, supporting different data movement patterns and runtime reconfiguration (Fig. 6(b)).

An A-Tile is surrounded by tile memory on each side, which also serves as inter-tile shared memory (Fig. 7(a)). Each tile memory consists of 8 banks for the systolic interconnects (systolic banks) and 8 banks for the router network (router banks). Each row or column of the 8x8 PE array in the A-Tile has its dedicated systolic bank and router bank for write access, and read access to 4 banks (2 systolic banks and 2 router banks) is shared by two adjacent arrays for inter-array data movement. Adjacent A-Tiles communicate via the shared tile memory or direct bypass connections, both providing 96GB/s bidirectional bandwidth. The four A-Tiles in a quadrant are interconnected by an AXI bus (Fig. 7(b)). Each A-Tile has a direct-memory-access (DMA) module for low-latency data transfer among tile memories via the AXI bus, allowing seamless data streaming between A-Tiles. EVA's mixed tile architecture and versatile interconnect fabric reduce runtime reconfiguration latency by at least 47x compared to Amber [1]. Typically, EVA adapts even faster through scheduling by C-Tiles, without requiring reconfiguration during runtime.

Flexible Workload Mapping and Optimization

A workload is composed of kernels, which are built from primitives. Primitives, like MACs or FFT butterflies, are mapped to one or multiple PEs. Kernels, such as matrix-vector multiplication (MVM) (Fig. 8(a)) or FFT (Fig. 8(b)), are assembled from these primitives within an A-Tile, across multiple A-Tiles or quadrants using EVA's interconnects. The modular composition allows for kernel and primitive reuse and quick adaptation as workloads evolve.

Workloads can be mapped spatially or temporally. For example, for a multi-antenna wireless DSP, FFT and MVM kernels can be spatially mapped to two A-Tiles and pipelined with C-Tile scheduling (Fig. 9(a)). The optimal mapping balances kernel latencies through suitable parallelization of each kernel, resulting in up to 2.95x performance improvement. Alternatively, workloads can be temporally mapped through runtime reconfiguration. For instance, for a visual object tracker, FFT, convolution, matrix transpose, and IFFT kernels are sequentially loaded to an A-Tile (Fig. 9(b)). This temporal mapping reduces hardware footprint and data movement, lowering energy consumption by 1.66x at the cost of a 24% increase in latency due to reconfiguration. We have developed a compiler (Fig. 9(c)) to explore mapping options and generate the A-Tile PE instructions and C-Tile scheduling and reconfiguration code.

Chip Measurement Results

EVA was fabricated in a 16mm² test chip in Intel16 (Fig. 10). Common kernels and example ML and DSP workloads were mapped to EVA, with results compared to Amber [1] (20mm² in 16nm) and DAP [2] (21mm² in 12nm) in Table I. EVA outperforms both Amber [1] and DAP [2] in performance. Table II compares EVA to recent state-of-the-art CGRA, array processors, and SoCs. EVA demonstrates the highest performance of 1.54TFLOPS and a compute density of 96.5GFLOPS/mm² at 0.9V and 1.5GHz, along with a competitive energy efficiency of 708GFLOPS/W at 0.49V and 111MHz. EVA offers the fastest reconfiguration latency of 75ns at 1.5GHz.

Acknowledgements

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References

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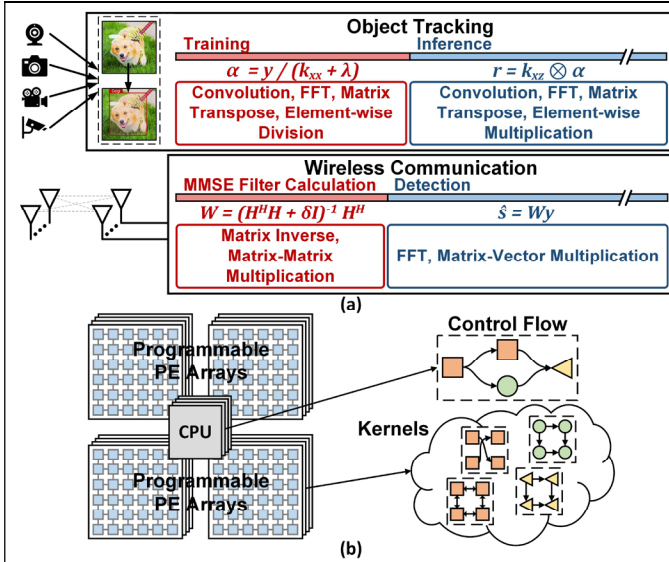


Fig. 1. (a) Edge computing tasks are composed of diverse compute kernels. (b) EVA, a flexible accelerator, supports runtime adaptation while maintaining performance and efficiency.

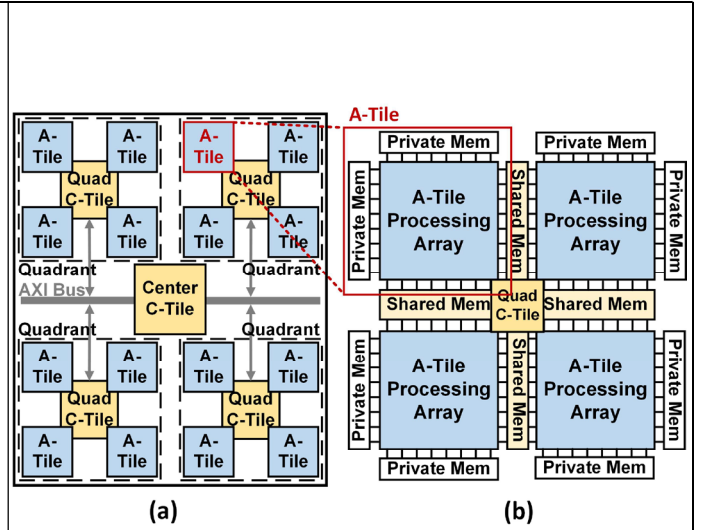


Fig. 2. (a) EVA is hierarchically constructed with 4 quadrants connected to a center C-Tile (an embedded RISC-V CPU core). (b) A quadrant contains a quadrant C-Tile and 4 A-Tiles. Each A-Tile has access to 24KB of memory.

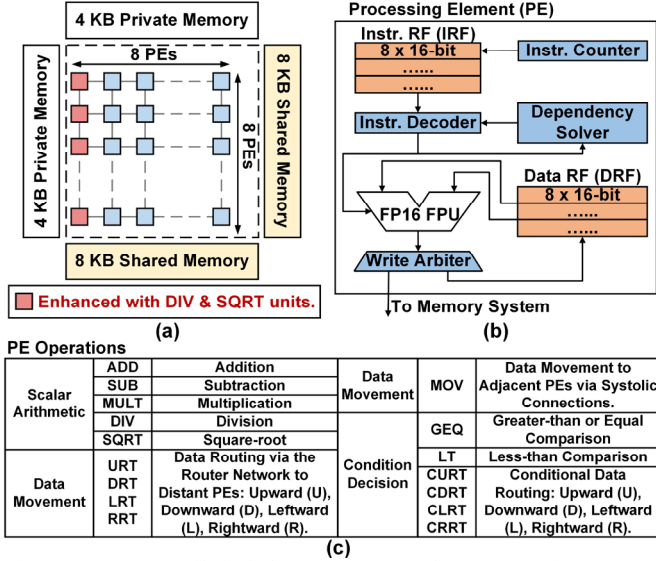


Fig. 3. (a) EVA's A-Tile includes a 64-PE array. (b) A PE contains an 8-entry Instruction RF (IRF). (c) PE operations include scalar computation, data movement, and conditional decisions.

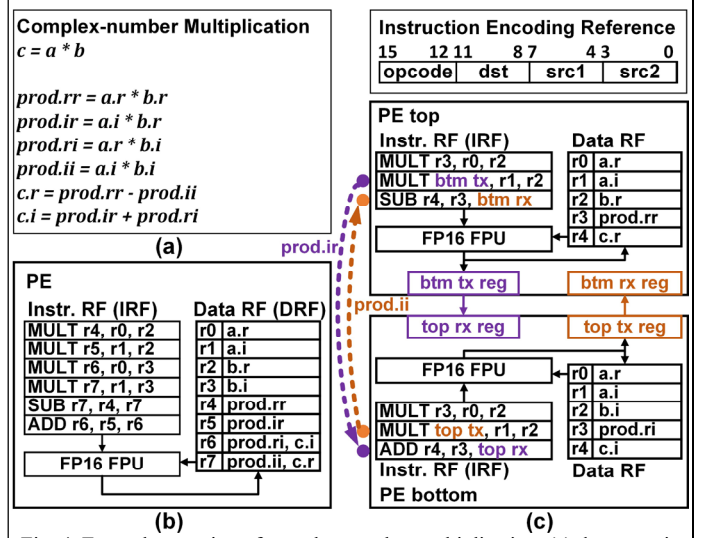


Fig. 4. Example mapping of complex-number multiplication: (a) decomposition of a complex-number multiplication into scalar operations; (b) mapping on a single PE for improved efficiency with local data access; and (c) mapping across multiple PEs for enhanced throughput.

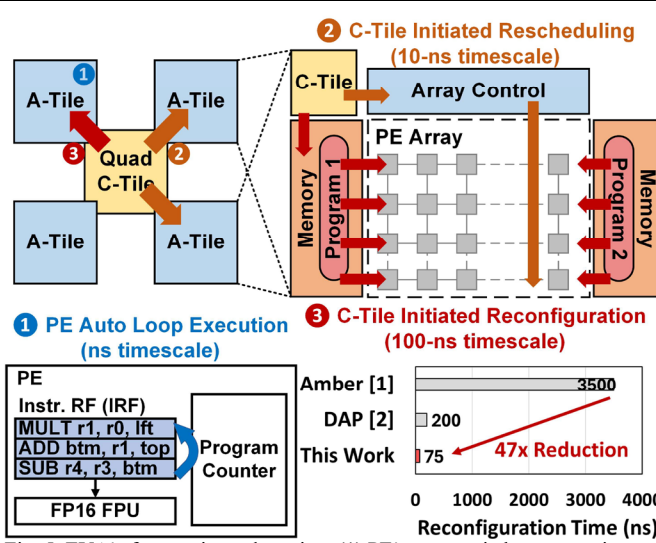


Fig. 5. EVA's fast runtime adaptation: (1) PE's automatic loop execution, (2) rescheduling by the quadrant C-Tile, and (3) reconfiguration by the quadrant C-Tile. Each quadrant is independently controlled by its own quadrant C-Tiles. Runtime reconfiguration latency is reduced compared to prior work.

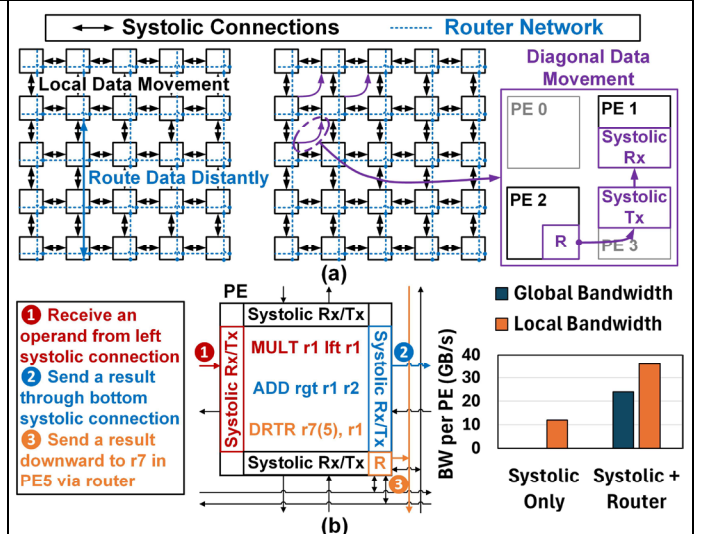


Fig. 6. EVA's interconnect fabric: (a) systolic interconnects and a router network facilitate PE communication; diagonal data movement is assisted by router; (b) an example illustrating data movement controlled by PE instructions. The interconnects provide enhanced bandwidth per PE.

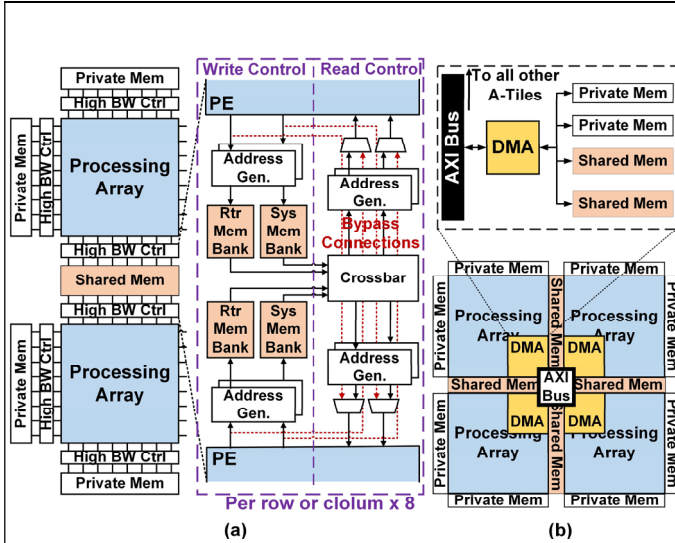


Fig. 7. A quadrant's memory system: (a) an A-Tile is surrounded by banked tile memory on each side, which also serves as shared memory between tiles; (b) A-Tiles are interconnected via an AXI bus. An A-Tile uses a direct-memory-access (DMA) module for low-latency data transfer.

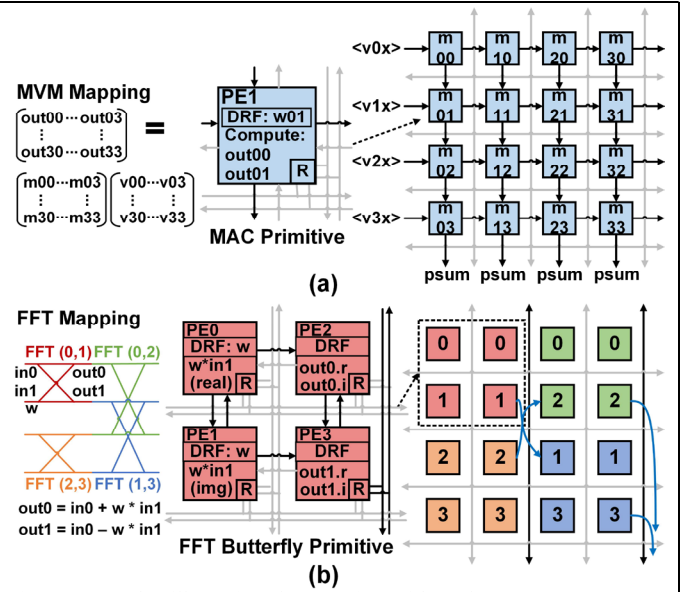


Fig. 8. Examples illustrating the mapping of kernels onto PE arrays: (a) matrix-vector multiplication (MVM); and (b) FFT.

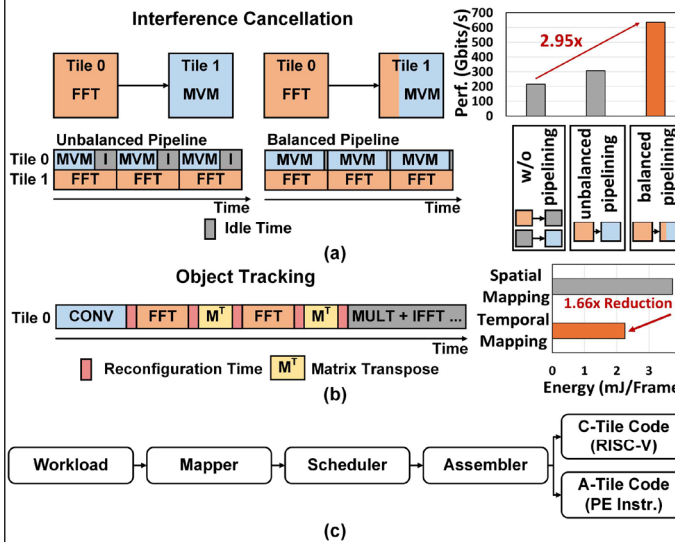


Fig. 9. Examples illustrating the mapping of workloads on EVA: (a) spatial mapping of interference cancellation for wireless communications; (b) temporal mapping of a visual object tracker; (c) EVA's compilation framework.

TABLE II. Comparison with state-of-the-art flexible processors

	EVA	Amber[1]	DAP[2]	ONYX[3]	SMIV[4]
Architecture	Tile-based Accelerator	SoC with CGRA	Systolic Array Processor	SoC with CGRA	SoC with FPGA
Technology (nm)	16	16	12	12	16
Area (mm ²)	16	20.1	21.16	23	25
Memory	300 KB	4.5 MB	522 KB	4.5 MB	9.025 MB
Data Format	FP16	BF16, INT16	FP16	BF16, INT16	FP16, INT16
Reconfiguration Time	75 ns (@1.5GHz)	3.5 us	~ 200 ns	N/A	0.45 ms (eFPGA)
Frequency (MHz)	111, 1500	955	8.55 - 506	970	> 1000
Performance (GFLOPS)	114 (@0.49V), 1544 (@0.9V)	342 (BF16) (@1.29V)	528 (@1.0V)	571 (INT16) (@0.78V)	10 - 100
Efficiency (GFLOPS/W)	708 (@0.49V), 196 (@0.9V)	483.3 (BF16) (@0.86V)	1014 (@0.42V)	756 (INT16) (@0.66V)	312.4 (eFPGA)
Compute Density (GFLOPS/mm ²)	96.5 (@1.5GHz, 0.9V)	17.0	24.9	24.8	13.0 (eFPGA)

TABLE I. Performance evaluation of kernels and workloads

	This Work (16 mm ² in 16 nm)	Amber[1] (20.1 mm ² in 16 nm)	DAP[2] (21.16 mm ² in 12 nm)
GeMM	567.57 GMAC/s	---	148.1 GMAC/s
FFT (256-pt)	44.21 MFFT/s	---	4.41 MFFT/s
CORDIC	1.09 GSamples/s	---	---
LDL Matrix Decomposition (4x4 matrix)	601.35 MMatrices/s	---	---
ResNet-18	445.94 GMAC/s	136 GMAC/s*	---
Object Tracking (64x64 frame)	883 Frames/s	---	---
Wireless Communication (4x4 matrix)	24.38 MMatrices/s	---	1.95 MMatrices/s
MMSE Filter Calculation	635.13 Gbits/s	---	213.12 Gbits/s
Detection	---	---	---

*Derived from latency and energy by layers.

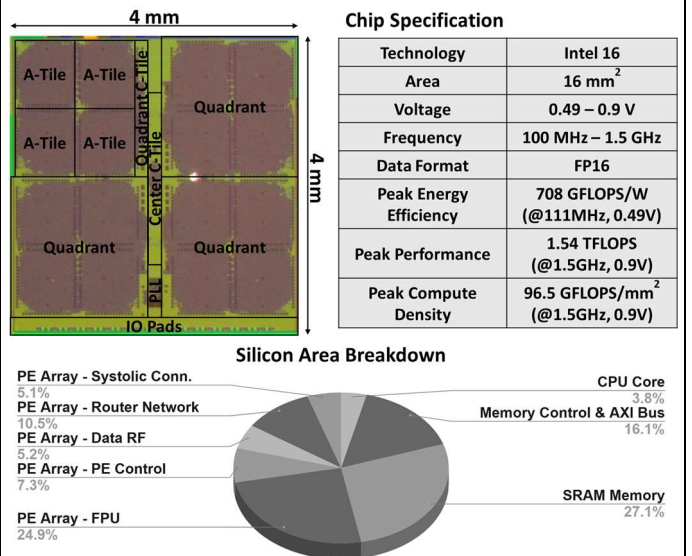


Fig. 10. Chip photo and summary.